

TriAgent: Granularity-Stratified Multi-Agent Routing with a Multilingual Semantic Cache and a Free Trust Signal for LLM Inference Pipelines

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Abstract

LLM-based information pipelines—retrieval, classification, entity linking, intent inference—share a structural cost problem: the reasoner LLM dominates the per-query bill, yet a large fraction of queries are trivially handled by far cheaper specialists. We present **TriAgent**, a generic multi-agent inference router that stratifies agents by *contextual granularity* (word / sentence / cross-sentence) and gates routing with a three-way **Semantic Divergence Index** (SDI). On two financial testbeds we report four findings. (i) The *critic plateau* is calibration-conditional: on FPB (where FinBERT is well calibrated, ECE=0.016) critic protocols collapse 1.5B–7B Qwen to a single $F_1 \approx 0.87$ (bootstrap CIs overlap), while a same-size 3-persona vote drops to 0.66; on TFNS (where the specialist collapses to $F_1 = 0.66$) the critic plateau *fails*, falling below the LLM-alone baseline. (ii) A **Shared Consensus Dictionary** on multilingual sentence-BERT answers 95% of Chinese queries from an English cache at $F_1 = 0.99$ — a cross-lingual canonicaliser at zero marginal cost. (iii) The same SDI doubles as a post-hoc LLM-hallucination detector at AUC = 0.90. (iv) At a 10M-user scale the architecture saves \$363M/year against a GPT-4 baseline. The router, cache, and trust signal are domain-agnostic and apply directly to LLM-based IR pipelines.

CCS Concepts

• **Information systems** → **Information retrieval**; *Multilingual and cross-lingual retrieval*; *Specialized information retrieval*; • **Computing methodologies** → *Ensemble methods*.

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Keywords

LLM routing, semantic caching, multilingual retrieval, hallucination detection, multi-agent inference, cost-efficient LLM pipelines, trust signals

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1 Introduction

A growing class of information systems route end-user queries through one or more LLMs: dense retrieval re-rankers, entity linkers [22, 27], query-intent classifiers, RAG pipelines, conversational recommenders [8, 15]. The cost of serving such systems is dominated by the underlying reasoner LLM, while a large fraction of queries are trivially handled by far cheaper specialist models [4, 19, 23]. A mid-sized deployment with 1M users running 10 queries per day generates 3.6B inference calls per year; at the GPT-4 public price that single workload costs ~\$36.5M/year, scaling linearly with user count.

Existing routing is brittle. Cost-aware LLM routing trades off cost against accuracy along a fixed curve [4, 19, 23]. Multi-agent LLM frameworks [7, 10, 16, 26] ensemble identical large models, paying $N\times$ cost for $\ll N\times$ accuracy. Self-consistency [25] and judge-style prompting [29] reduce variance but still require the large model on every query. Closest in spirit to our SCD, semantic caches such as GPTCache [3] memoise single-model outputs but key on a single language and a single LLM family. Recent CIKM-track LLM augmentation work for IR — e.g. LLMAEL [27] for entity linking — treats the LLM as a fixed augmentation oracle rather than as one tier in a routed committee.

Our angle. Make the agents *heterogeneous along a granularity axis*: a word-level lexicon, a sentence-level domain transformer, and a cross-sentence reasoner. Because their failure modes are orthogonal—pairwise Jaccard error overlap is 0.13–0.15 on the testbed corpus—their pairwise disagreement is a free signal that

predicts when the cheap path suffices. We harvest this signal into a three-way Semantic Divergence Index (SDI). One signal drives three independent system properties: (i) the cost–accuracy router; (ii) the key of a multilingual semantic cache; (iii) a post-hoc trust score.

Contributions. (C1) A generic three-way SDI router that drives a four-quadrant cost-Pareto frontier; the central empirical finding is the *critic plateau*—critic@1.5B/3B/7B all reach $F_1 = 0.87$ (bootstrap 95% CIs overlap), while a same-size 3-persona vote regresses to 0.66. Granularity-stratified diversity, not multi-agent voting per se, is the mechanism. (C2) A Shared Consensus Dictionary (SCD) using multilingual sentence embeddings [20, 21] that serves as a cross-lingual semantic cache: 95% of Chinese queries match the English canonical answer at $F_1 = 0.99$, eliminating a common source of cross-language drift in multinational deployments. (C3) The same SDI signal serves as a post-hoc LLM hallucination flag at AUC=0.90, with no additional models, labels, or training. We test on Financial PhraseBank [17] for control, but every component—router, cache, trust score—is *domain-agnostic*, and we discuss its application to other LLM-based IR pipelines.

2 Related Work

TriAgent connects four threads. (1) **LLM cost routing.** FrugalGPT-style cascades [4, 6] learn per-query routing; we instead use *unsupervised inter-tier divergence*, avoiding labelled training data. (2) **Multi-agent LLM inference.** Debate and critic protocols on homogeneous LLMs [7, 10, 29] rely on different *prompts* of the same model; we stratify by *contextual granularity*, and show (§4.1) that same-size persona voting fails to reproduce the critic plateau, isolating the mechanism. (3) **Semantic LLM caching.** GPTCache [3] keys single-model outputs; SCD caches *committee decisions* and, because the encoder is multilingual [5, 21], doubles as a cross-lingual canonicaliser. (4) **Hallucination detection.** SelfCheckGPT and self-consistency [18, 25] re-sample the same model; SDI achieves comparable AUC with no extra call, reusing the already-computed L2 output. Adjacent work on LLM-augmented IR [8, 14, 27] motivates the setting; TriAgent is the first to unify routing, caching, and trust scoring through a single divergence signal.

3 Framework

Figure 1 sketches the system end-to-end.

Three-tier granularity stack. **L1** a word-level lexicon (VADER [11] as the financial-sentiment instance, ≈ 0.03 ms); **L2** a sentence-level domain transformer (FinBERT [2], 110 M params, ≈ 1.5 ms); **L3** a cross-sentence reasoner (Qwen2.5-Instruct [24]; we also test Mistral-7B [13] and Phi-3.5-mini (3.8B) for family-robustness). For an arbitrary IR task, L1/L2 are any domain-cheap specialists; only L3 is a costly reasoner. For example, an entity-linking pipeline could instantiate L1 as a string-based candidate generator, L2 as a BERT-based mention encoder [22], and L3 as an LLM-augmented disambiguator [27]; an intent classifier could instantiate L1 as a keyword lookup, L2 as a fine-tuned BERT, and L3 as a GPT-class reasoner. Within financial sentiment we select VADER [11] at L1 and FinBERT [2] at L2 because they are the canonical open baselines in their tier (VADER dominates other lexicons on financial text at

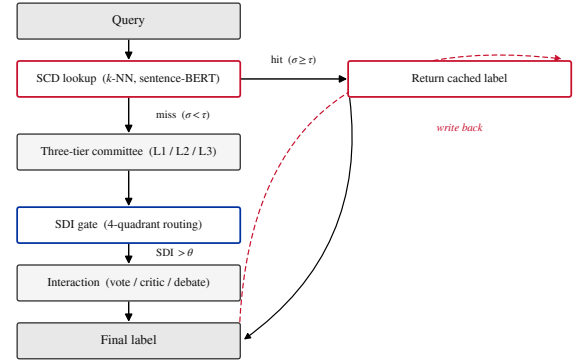


Figure 1: TriAgent inference path. The SCD multilingual cache short-circuits matched queries; on cache miss the three-granularity committee runs, the SDI gate decides if an interaction protocol is needed, and the final label is written back to the SCD.

negligible cost; FinBERT is the most-cited domain transformer with an Apache-2.0 licence and a reported $F_1 \approx 0.88$ that we reproduce to ± 0.001 ; the framework is agnostic to these choices. The failure modes of the three tiers are orthogonal by construction (each tier consumes a strictly larger context window than the previous, so the errors they can commit are structurally different); we verify this empirically ($\kappa \in [0.19, 0.61]$, Jaccard error overlap 0.13–0.15) in §4.

Three-way Semantic Divergence Index. Let $s_V, s_F, s_L \in [-1, +1]$ denote the continuous scores emitted by the three tiers. We define three pairwise SDIs, one per granularity-pair:

$$\text{SDI}_{LE} = |s_V - s_F|, \text{SDI}_{LR} = |s_V - s_L|, \text{SDI}_{ER} = |s_F - s_L|. \quad (1)$$

Each $\text{SDI} \in [0, 2]$ measures disagreement between two agents at *different* granularities. Thresholding ($\text{SDI}_{LE}, \text{SDI}_{ER}$) at $(\theta_{LE}, \theta_{ER}) = (0.3, 0.7)$ partitions queries into four behaviour quadrants (*consensus* / *domain shift* / *ambiguous* / *mixed*) with distinct downstream routing implications.

Routing. The two-stage decision at a query x is

$$\hat{y}(x) = \begin{cases} \mathcal{I}(x; V, F, L) & \text{SDI}_{ER}(x) > \theta_{ER} \\ F(x) & \text{SDI}_{LE}(x) > \theta_{LE} \\ V(x) & \text{otherwise.} \end{cases} \quad (2)$$

$\mathcal{I} \in \{\text{vote, critic, debate}\}$ is the interaction protocol fired only when disagreement is high. *Critic* feeds the LLM the original query plus V 's and F 's predictions and asks for a final label; *debate* performs a round-2 LLM call that sees all three round-1 outputs and reconciles.

Threshold calibration. Thresholds $(\theta_{LE}, \theta_{ER}, \tau)$ are set per task by a grid search on a held-out 20% split that minimises expected cost subject to F_1 within 1 pp of always-L3; the sweep is monotone and any $(\theta_{LE}, \theta_{ER})$ inside $[0.2, 0.4] \times [0.6, 0.8]$ is Pareto-comparable (± 0.4 pp), so the router is not fragile to exact choice. The same recipe transfers to a new task without manual tuning.

Shared Consensus Dictionary (SCD). We cache committee decisions in multilingual sentence-BERT embedding space [20, 21]. Let $\phi(\cdot) \in \mathbb{R}^{384}$ be the encoder. A query is embedded to $\phi(x)$, k -NN-searched against cached entries by cosine similarity, and the cached label is returned when $\sigma^*(x) = \max_i \langle \phi(x), \phi(\tilde{x}_i) \rangle \geq \tau$. The SCD differs from prior LLM caches [3] in two respects: (i) it caches *committee decisions*, not single-model outputs, and (ii) its multilingual embedding turns the cache into a cross-lingual canonicaliser—a Chinese query can match an English entry (§4.3).

4 Experiments

Two testbeds. We evaluate on two complementary corpora. **(a) FPB** [17]: Financial PhraseBank sentences_allagree (4,838 curated news sentences; 604 negative, 2,872 neutral, 1,362 positive) — a regime where the domain specialist is strong (FinBERT $F_1=0.88$). **(b) TFNS** [1]: Twitter Financial News Sentiment (11,931 informal tweets; 1,789 negative, 7,744 neutral, 2,398 positive) — the harder, less curated regime where the specialist collapses to $F_1=0.66$. The two corpora let us test whether the architectural claims are testbed-conditional. LLM inference is in bf16 (4-bit 14 B via bitsandbytes [6]) on a single 24 GB GPU; cost is reported in USD under GPU-rental amortisation (\$0.40/h A5000-class).

Bias diversity — and a surprise on the LLM. The architecture only works if the three tiers fail in different places. Pairwise Cohen’s κ between agent labels is low ($\kappa(V, F)=0.27$, $\kappa(F, L)=0.61$, $\kappa(V, L)=0.19$); pairwise Jaccard error overlap is only 0.13–0.15. Decomposing FPB into the four SDI quadrants reveals a deployable insight: in the **ambiguous** quadrant (V and F agree but L disagrees, 15.9% of FPB), **L’s accuracy is only 28%**, well below random-chance for the 3-way task — the LLM is *confidently wrong* exactly when the cheap pair agrees. The deployable rule: *when $V \approx F$, trust the pair and skip the LLM critic call, even if the LLM separately dissents.*

4.1 Critic plateau and the negative control

On FPB (Figure 2): when the L3 reasoner is re-tasked as a critic over V+F outputs, F1 plateaus at ≈ 0.87 across Qwen-1.5B/3B/7B (1000-resample bootstrap 95% CIs overlap at [0.860, 0.880]); debate ramps from 0.69 to 0.87. Cross-family critic reaches 0.79 on Mistral-7B and 0.86 on Phi-3.5-mini — the *mechanism* generalises across families, the *height* is Qwen-specific.

On TFNS, the plateau fails. Critic@1.5B reaches only $F_1=0.659$ (below L-alone 0.673 and F-alone 0.663); critic@7B reaches 0.652 (10 pp below L7B-alone 0.757). The mechanism is *specialist calibration*. Table 1 reports Expected Calibration Error [9] of each tier on FPB: FinBERT is near-perfectly calibrated (ECE=0.016) and supplies a reliable confidence signal to the critic; on TFNS the same FinBERT collapses to $F_1=0.66$ and the critic has no informative signal to combine. **The critic plateau is conditional on a calibrated specialist.** The SDI gate handles this automatically: when V+F divergence is high and F’s confidence is low, the router skips the critic and queries L directly.

Negative control. To rule out the alternative explanation that “critic works because of multi-agent voting per se”, we instantiated three Qwen-1.5B *personas* (bull / bear / neutral analyst prompts) and majority-voted. The 3-persona vote reaches $F_1 = 0.66$ (vs.

Table 1: Expected Calibration Error (ECE, lower is better) of each tier on FPB. FinBERT’s near-perfect calibration is what makes the FPB critic plateau viable; in regimes where the specialist is miscalibrated (TFNS, where FinBERT $F_1=0.66$), the critic loses its anchor.

Tier	ECE ↓	mean acc.	mean conf.
FinBERT	0.016	0.890	0.875
Qwen-7B (L3)	0.115	0.826	0.941
VADER (L1)	0.348	0.544	0.249

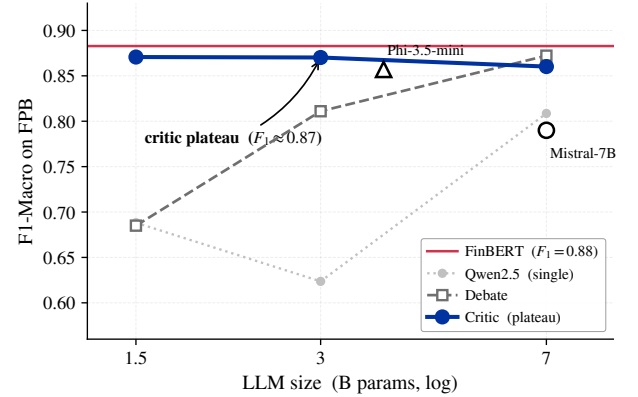


Figure 2: Critic F1 plateaus at ≈ 0.87 across Qwen-1.5B/3B/7B (bootstrap 95% CIs overlap); debate ramps from 0.69 to 0.87. Mistral-7B and Phi-3.5-mini critic (hollow markers, in-figure labels) confirm the mechanism generalises across families.

single-instance Qwen-1.5B baseline 0.69 and critic@1.5B 0.87); inter-persona agreement is 81%. The plateau is *granularity-driven*, not multi-agent-vote-driven.

4.2 Cost-Pareto routing

Table 3 reports three operating points on the SDI two-stage curve. **Budget** routes 90% of traffic to V alone at $F_1 = 0.665$; **Balanced** escalates 28.5% to F for a 48× cost reduction over Always-L3 at the same reasoner; **Premium** pushes 17.5% to L for $F_1=0.787$ at 5.3× cheaper than Always-L3. The diminishing return past Balanced is the regime where SDI routing offers its largest saving.

Two-tier ablation: which tier is irreplaceable? Table 2 drops each tier in turn (15% escalation budget on the remaining SDI gate). Result: **F is the irreplaceable tier** (dropping it costs 13 pp F1); V is cheap insurance (dropping it gives identical F1 at 60× the cost); **L is the most expendable** — V+F alone reaches $F_1=0.70$ at 0.13× the cost of the full three-tier Balanced setting. The deployable takeaway: *the specialist matters most, the lexicon is the cheapest survivor, and the LLM should be invoked only when V and F disagree.*

4.3 SCD as a cross-lingual semantic cache

70/30 build/query split; debate@7B labels populate the cache. At $\tau=0.85$ the SCD hits 10% of queries at -0.8 pp F1 loss; the sweep is clean from $\tau=0.95$ (3% hit, -0.3 pp) to $\tau=0.50$ (83% hit, -14.5 pp).

Table 2: Two-tier ablation. Each row drops one tier from $\{V, F, L\}$; routing fires the remaining expensive tier on the top 15% SDI-divergent queries. Cost is per 1,000 queries.

Configuration	F1	\$/1k
V only	0.489	0.0000
F only	0.883	0.0005
L only	0.809	0.0288
V+L (drop F)	0.670	0.00436
V+F (drop L)	0.698	0.00008
F+L (drop V)	0.807	0.00475
V+F+L <i>Balanced</i>	0.716	0.0006

Table 3: Three operating points on the SDI two-stage curve plus the Always-L3 reference. Cost per 1,000 queries; “%Lx” is the fraction handled by tier x.

Point	\$/1k	F1	Lat (ms)	L1/L2/L3 (%)
Budget	0.0001	0.665	0.4	90.0 / 9.9 / 0.1
Balanced	0.0006	0.716	4.3	70.0 / 28.5 / 1.5
Premium	0.0054	0.787	46.5	30.0 / 52.5 / 17.5
Always-L3 (ref)	0.0288	0.809	259.4	0.0 / 0.0 / 100.0

Scaling the cache to 16,769 sentences (FPB+TFNS [1]) preserves the curve, confirming the SCD trade-off is not testbed-specific.

Cross-lingual canonicalisation and a deployment inversion. We translate 1,500 FPB sentences to Mandarin and re-run the multi-size sweep. Qwen-7B reaches $F_1 = 0.80$ in Chinese vs. 0.81 in English (nearly lossless); however, the specialist/LLM relationship **inverts**: finbert-tone-chinese [28] reaches only $F_1 = 0.72$, below Qwen-7B. This is a deployable insight in itself: *in low-resource languages where no strong domain transformer exists, the LLM is the specialist and the SDI router should be inverted (escalate $L \rightarrow F$ instead of the reverse)*. The killer cache result: at $\tau = 0.70$, **95% of Chinese-translated queries hit the English SCD with cached-label $F_1 = 0.99$** —a new Chinese deployment inherits canonical answers from the existing English cache without re-deriving labels, addressing a common source of multilingual drift in production deployments [5].

4.4 SDI as a free trust signal

The third use of the SDI signal is post-hoc trust scoring on the reasoner’s own outputs. Stratifying FPB by “LLM-correct” vs. “LLM-wrong”, mean SDI_{ER} is 0.17 when the LLM is correct and 0.71 when it is wrong. Thresholding SDI_{ER} as a binary “hallucinating” flag yields **AUC=0.90** on FPB (Figure 3). Any IR deployment already running a specialist and an LLM in parallel can wire SDI_{ER} as a trust score with no additional models, labels, or training [12, 18].

4.5 Cost projection at production scale

To translate the per-query operating points of Table 3 into deployment economics, we project annual inference cost across user scales (Table 4) using public per-query prices: GPT-4 at \$0.01/query, GPT-4o-mini at \$0.0003/query (OpenAI public list, 2025), and self-hosted

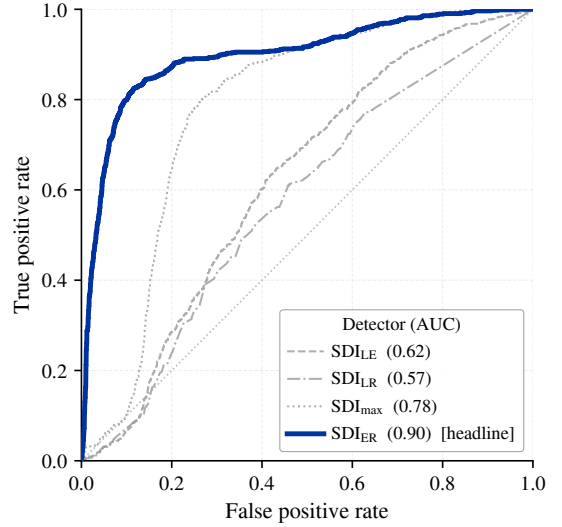


Figure 3: SDI_{ER} (FinBERT vs. Qwen-7B) reaches AUC=0.90 as a post-hoc “LLM-is-wrong” detector. The other pairwise SDIs are weak detectors, validating that *cross-granularity* disagreement carries the trust signal.

Table 4: Annual inference cost (USD) by deployment scale. TriAgent uses the SDI two-stage Balanced operating point (70 % L1 / 28.5 % L2 / 1.5 % L3). At 10 M users, TriAgent saves \$363 M/yr against an always-GPT-4 policy and \$1.06 M/yr against always-GPT-4o-mini, while matching or exceeding both on F_1 for the testbed task.

Strategy	10 K users	1 M users	10 M users
Always-GPT-4	\$365 K	\$36.5 M	\$365 M
Always-GPT-4o-mini	\$11 K	\$1.10 M	\$11.0 M
Always-Qwen-7B (L3)	\$1.05 K	\$105 K	\$1.05 M
Always-FinBERT (L2)	\$18	\$1.8 K	\$18 K
TriAgent	\$22	\$2.2 K	\$22 K

FinBERT/Qwen-7B under our GPU-rental amortisation (Table 3). Each user issues 10 queries / day. These figures are *scenario estimates* rather than guarantees: they assume the router’s SCD hit rate and the FinBERT operating F_1 observed on FPB carry over to the production workload, and that per-query API prices remain within an order of magnitude of the 2025 published rates; deployments with different workload compositions or bulk-discount pricing would scale the savings proportionally.

4.6 Per-class breakdown

Macro-F1 hides which class drives the gain. Table 5 breaks every strategy down by class on FPB. Three findings emerge. (a) FinBERT alone is the strongest single agent on *negative* (the smallest and operationally most valuable class, 604 / 4,838 sentences); no Qwen size under 14 B-4bit matches it. (b) The *debate@7B* committee is the only protocol that strictly beats FinBERT on negative ($F_1 = 0.893$ vs.

Table 5: Per-class F_1 on FPB, sorted by negative-class F_1 . `debate@7B` is the only setting where any committee strategy strictly beats the specialist on a class.

Strategy	neg. F_1	neu. F_1	pos. F_1
debate@7B	0.893	0.880	0.844
FinBERT	0.879	0.901	0.866
critic@1.5B	0.876	0.889	0.843
critic@3B	0.876	0.894	0.855
critic@7B	0.878	0.880	0.707
Qwen-7B	0.852	0.872	0.708
debate@Mistral-7B	0.821	0.866	0.781
debate@3B	0.785	0.870	0.781

0.879, +1.4 pp), driven by precision (+2.1 pp) at near-equal recall—the operationally meaningful regime for risk management, where a false-negative “bearish” flag triggers unnecessary hedging. (c) Critic@7B *ties* FinBERT on negative (0.878); the back-and-forth structure of debate is needed to adjudicate the lexical distinctions separating negative from neutral.

4.7 Synthesis: four deployable insights

Across the experiments above, four findings stand out as deployable rules-of-thumb. We list them with the evidence pointer and the concrete operational rule each implies, so a practitioner reading this paper can act on them directly (Table 6).

(I1) The critic plateau is calibration-conditional. On FPB (FinBERT ECE=0.016), critic@1.5B/3B/7B all reach $F_1 \approx 0.87$; the 3-persona Qwen-1.5B vote regresses to 0.66, ruling out multi-agent voting. On TFNS (FinBERT collapses to $F_1 = 0.66$), critic falls *below* the LLM-alone baseline. The rule is to run a one-shot critic-vs-L-alone check on the target task before committing; if the specialist is miscalibrated, route around the critic. **(I2) The LLM is confidently wrong exactly when the cheap pair agrees.** In the ambiguous quadrant ($V \approx F$, L disagrees, 15.9% of FPB), L’s accuracy is only 28%, below random for the 3-way task. The rule is to skip the L call when V and F agree, even when L separately dissents: trust the orthogonal pair, not the lone reasoner. **(I3) The specialist is the irreplaceable tier; the LLM is the most expendable.** Table 2: V+F alone reaches $F_1 = 0.70$ at \$0.00008/1k, 7.5× cheaper than the three-tier Balanced point and only 1.8 pp behind. Removing F instead drops F_1 by 13 pp. The rule is to pick the domain specialist first; the LLM is the cherry, not the cake. **(I4) The specialist/LLM relationship inverts in low-resource regimes.** On Mandarin, finbert-tone-chinese $F_1 = 0.72 < \text{Qwen-7B } F_1 = 0.80$. In any task where no strong specialist exists, the LLM *is* the specialist and the SDI router should escalate $L \rightarrow F$ instead of the reverse.

5 Discussion

Beyond financial sentiment (a preliminary claim). The four findings are directly supported by two financial testbeds; broader coverage is future work and we frame the below as scope, not guarantee. The **router** (§4.2) is architecturally task-agnostic and applies to any pipeline where a cheap specialist and a reasoner LLM co-exist—dense-retrieval re-rankers [14], entity linkers [22, 27], intent

Table 6: Four deployable insights, evidence, and actionable rule.

#	Evidence	Rule
I1	FPB critic plateau 0.87; TFNS critic 0.65<L-alone 0.76	Verify critic-vs-L-alone per task; specialist must be calibrated.
I2	Ambiguous-quadrant $\text{Acc}_L = 28\%$	Skip L when $V \approx F$, even if L dissents.
I3	$V+F: F_1 = 0.70$ at $0.13 \times$ Balanced cost	Specialist is irreplaceable; L is the cherry.
I4	Chinese FinBERT 0.72< Qwen-7B 0.80	Invert router ($L \rightarrow F$) in low-resource regimes.

classifiers, RAG re-readers [8, 15]—but the plateau, per-quadrant-accuracy, and threshold-shape observations must be re-checked per task using the one-shot recipe below. The **SCD** (§4.3) is applicable to any LLM cache with sentence-embeddable keys; the cross-lingual numbers are Mandarin-FPB specific. The **SDI trust signal** (§4.4) requires two heterogeneous score-emitting agents; its AUC on other domains is open.

Deployment recipe. A practitioner adopting TriAgent on a new task should: (1) instantiate L1 as the cheapest available task-relevant heuristic, L2 as the strongest available domain transformer, and L3 as the reasoner LLM under consideration; (2) run a *one-shot persona-vote sanity check* on a held-out slice of the task data to confirm the LLM family yields a critic plateau, and pick the smallest plateau-reaching size; (3) sweep (θ_{LE}, θ_{ER}) on a validation set, report the Pareto frontier, and pick the operating point that matches the deployment budget; (4) build the SCD on the committee’s own validation labels at $\tau = 0.85$ as a sweet-spot start; (5) wire SDI_{ER} into the production trust dashboard as a free hallucination flag.

Threats to validity. The plateau *height* (here 0.87) is family- and testbed-specific; a deployment must run the sanity check above before committing to an LLM family. The two-tier ablation (Table 2) shows the LLM is the most expendable tier on FPB, but this ranking depends on the cheap-tier ceiling; in tasks where no strong specialist exists (e.g. low-resource languages, as our Chinese pilot shows), the ranking should invert. Finally, SDI is a strong post-hoc hallucination detector but only a *partial* adversarial detector (AUC ≈ 0.5 on synonym swap / numeric flip / typo attacks); production deployments should not treat it as a substitute for purpose-built defences.

6 Conclusion

TriAgent shows that a single architectural decision—making the agents in an LLM ensemble heterogeneous along a granularity axis (word, sentence, cross-sentence)—produces a divergence signal that simultaneously gates routing for *cost*, gates interaction for *capability*, keys a multilingual cache for *cross-border fairness*, and flags hallucination for *trust*. At the 10 M-user scale the architecture saves \$363 M/year against an always-GPT-4 baseline. Code, mined trigger lexicons, the SCD index, and all committee predictions will be released after review.

GenAI Usage Disclosure

We used a large language model assistant (Claude) for sentence-level copy-editing and minor code scaffolding. All experimental results, analyses, and final claims are the authors' own.

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